## Session 23 — Definition and graph of the reciprocal function

Reciprocal function

15/12/2026 • B : 2 h – C : 1 h

### Objectives of the session

· Determine the domain.

· Draw the curve.

· Use symmetries.

### Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x + 5 - x + 2$. .....

2. Give the forbidden value of $\dfrac{x+1}{x}$. .....

3. Calculate $\dfrac{1}{5}$ and $\dfrac{1}{-5}$. .....

4. Differentiate 3 $x^2 - 5x + 1$. .....

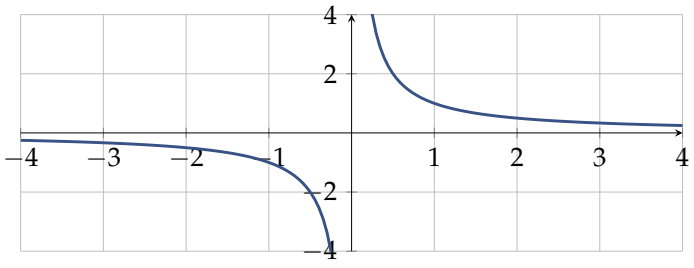
5. Solve $(x-2)(x+1) = 0$. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: determine the domain and draw the graph.

### Investigation

Calculate $\dfrac{1}{x}$ for $x = -4, -2, -1, -\dfrac{1}{2}, \dfrac{1}{2}$, 1, 2, 4. Plot the points and describe the symmetries. Why $x = 0$ missing?




### Written summary

The reciprocal function is $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. Its graph is a hyperbola symmetric about the origin.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

**Practice · essentials**

1. Give the domain and the asymptotes of $f(x) = 5 + \frac{7}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \frac{25}{x}$.

**References in the textbook**

**Extensions:** Calao pp. 18-21; ex. p. 24 #1-3. The exercise in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

**Calao textbook · p. 101, Exercise 42**

The aim of this exercise is to draw the graphs of the functions $f$ and $g$ defined on $\mathbb{R}^*$ by: $f(x) = \frac{2}{x}$ & $g(x) = 4 - \frac{3}{x}$.

1. Complete the table of values for $x \in \{-5; -4; -2; -0{,}5; -0{,}2; -0{,}1; 0{,}1; 0{,}2; 0{,}5; 1; 2; 4; 5\}$.
2. Using a table of values, draw the graphs of the functions $f$ and $g$.

**Independent work**

Redo practice exercise 2 with the following data: replace the initial value with 235 and the rate or common ratio with 7. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

## Session 24 — Behaviour at the endpoints and asymptotes

Reciprocal function

17/12/2026 • B : 1 h – C : 2 h

### Objectives of the session

· Describe the behaviour at the endpoints.

· Identify the asymptotes.

· Read a representation.

### Fluency · 6 to 8 minutes, without calculator

1. Calculate 5% of 120. .....

2. Calculate $\frac{1}{6}$ and $\frac{-1}{6}$. .....

3. Differentiate $x^2 - 5x + 1$. .....

4. Solve $(x-2)(x+1) = 0$. .....

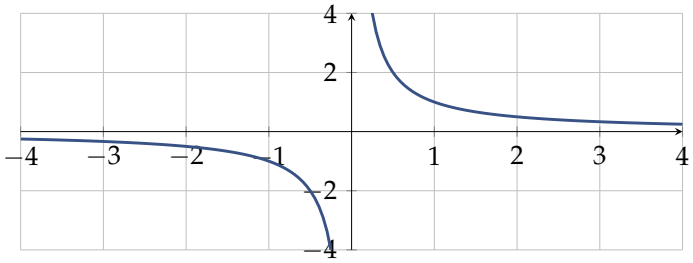
5. Give the forbidden value of $\frac{x+1}{x}$. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: describe the behaviour at the endpoints and identify the asymptotes.

### Investigation

Using a table of values close to 0 and for large values of $|x|$, describe $\frac{1}{x}$. Express each behaviour with an asymptote and a sentence.




### Written summary

For $f(x) = \frac{1}{x}$, $x = 0$ is a vertical asymptote and $y = 0$ is a horizontal asymptote. The branches approach them without crossing them.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears

in the teacher's edition.

**Practice · essentials**

1. Give the domain and the asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{36}{x}$.

**References in the textbook**

**Extensions:** Calao pp. 18-21; ex. p. 24 #1-3. The exercise in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

**Calao textbook · p. 95, Exercise 16**

We want to plot the graph of the function $g$ defined on $\mathbb{R}^*$ by: $g(x) = 3 - \dfrac{2}{x}$.

1. Complete a table of values for

$$x \in \{-1\,000; -500; -200; -100; -50; 50; 100; 200; 500; 1\,000\}.$$

   What can we deduce about the behaviour of $g$ at $-\infty$ and at $+\infty$ ?

2. Complete a table of values for $x = -0{,}5; -0{,}1; -0{,}01; 0{,}01; 0{,}1; 0{,}5$. What can we deduce about the behaviour of $g$ at 0?
3. Copy and complete the table of values from the textbook, then plot the curve of the function $g$.

**Independent work**

Redo practice exercise 3 with the following data: replace the initial value with 240 and the rate or common ratio with 2. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

## Session 25 — Derivative and monotonicity

Reciprocal function

05/01/2027 • B : 2 h – C : 1 h

### Objectives of the session

· Differentiate the reciprocal function.

· Justify the variations.

· Draw up a table.

### Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 10% increase

2. Differentiate 3 $x^2 - 5x + 1$. .....

3. Solve $(x-2)(x+1) = 0$. .....

4. Give the forbidden value of $\dfrac{x+1}{x}$. .....

5. Calculate $\dfrac{1}{7}$ and ~~$\dfrac{1}{7}$~~. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate the reciprocal function and justify the variations.

### Investigation

From the rate of change of $f(x) = \dfrac{1}{x}$, deduce $f'(x) = -\dfrac{1}{x^2}$. Study its sign and check consistency with the graph.

### Written summary

For $x \neq 0$, $f'(x) = -\dfrac{1}{x^2} < 0$. The reciprocal function decreases on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

### Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 2 + \dfrac{4}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{4}{x}$.

**References in the textbook**

**Extensions:** Calao pp. 20-21; ex. p. 24 #11-13, then p. 28 #43-44. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

**Calao textbook · p. 94, Exercise 3**

Calculate the derivatives of the following functions defined on $\mathbb{R}^*$.

a. $f(x) = 2x - 3 + \dfrac{4}{x}$

b. $g(x) = 3x - 5 - \dfrac{1}{x}$

c. $f(x) = 7x^3 - 2x + \dfrac{1}{x}$

d. $g(x) = 3x^2 - 5x - 2 + \dfrac{4}{x}$

**Independent work**

Redo practice exercise 2 with the following data: replace the initial value with 245 and the rate or common ratio with 3. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

## Session 26 — Combinations with polynomial functions

Reciprocal function

07/01/2027 • B : 1 h – C : 2 h

### Objectives of the session

· Differentiate a combination.

· Simplify an expression.

· Organise a study.

### Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x+8-x+2$. .....
2. Solve $(x-2)(x+1)=0$. .....
3. Give the forbidden value of $\dfrac{x+1}{x}$. .....
4. Calculate $\dfrac{1}{8}$ and $-\dfrac{1}{8}$. .....
5. Differentiate 3 $x^2-5x+1$. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate a combination and simplify an expression.

### Investigation

A company produces $x>0$ hundred objects and its average cost is $C(x)=2x+30+\dfrac{200}{x}$. Calculate $C'(x)$, factorise its numerator and find the production level that minimises the average cost.

### Written summary

To study $P(x)+\dfrac{k}{x}$, determine the domain, calculate the derivative, reduce to the same denominator, study its sign then draw up the variations.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

### Practice · essentials

1. Give the domain and the asymptotes of $f(x)=3+\dfrac{5}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0,+\infty[$ the function $g(x)=x+\dfrac{9}{x}$.

**References in the textbook**

**Extensions:** Calao, p. 96, No. 14-21. The exercise reproduced in this session is identified below. by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

**Calao textbook · p. 96, Exercise 20**

Let $f$ the function defined on $\mathbb{R}^*$ by $f(x) = 4x + 1 + \frac{9}{x}$.

1. Show that for every real $x$ non-zero: $f'(x) = \frac{(2x-3)(2x+3)}{x^2}$.
2. For every real $x$ non-zero, study the sign of $f'(x)$.
3. Deduce the variation of the function $f$.

**Independent work**

Redo practice exercise 3 with the following data: replace the initial value with 250 and the rate or common ratio with 4. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

## Session 27 — Sign and factorisation

Reciprocal function

12/01/2027 • B : 2 h – C : 1 h

### Objectives of the session

· Factorise.

· Build a sign table.

· Respect the forbidden values.

### Fluency · 6 to 8 minutes, without calculator

1. Calculate 20% of 180. .....

2. Give the forbidden value of $\dfrac{x+1}{x}$. .....

3. Calculate $\dfrac{1}{9}$ and $-\dfrac{1}{9}$. .....

4. Differentiate 3 $x^2 - 5x + 1$. .....

5. Solve $(x-2)(x+1) = 0$. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: factorise and build a sign table.

### Investigation

Study the sign of $\dfrac{(x-2)(x+3)}{x}$. Construct the table with the three critical values, distinguish zero and forbidden value, then solve the inequality $\dfrac{(x-2)(x+3)}{x} \geq 0$.

### Written summary

In a quotient, the sign depends on the numerator and the denominator. The forbidden values are excluded from the table, even if a factor formally cancels.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

### Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 4 + \dfrac{6}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{16}{x}$.

**References in the textbook**

**Extensions:** Calao, pp. 96-101. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

**Calao textbook · p. 96, Exercise 21**

Let $g$ the function defined on $\mathbb{R}^*$ by $g(x) = x + 10 - \frac{1}{x}$.

1. Show that for every real $x$ non-zero: $g'(x) = \frac{x^2+1}{x^2}$.
2. For every real $x$ non-zero, study the sign of $g'(x)$.
3. Deduce the variation of the function $g$.

**Independent work**

Redo practice exercise 2 with the following data: replace the initial value with 255 and the rate or common ratio with 5. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

## Session 28 — Derivatives of polynomials of degree at most 3

Reciprocal function

14/01/2027 • B : 1 h – C : 2 h

### Objectives of the session

· Differentiate a polynomial.

· Study a derivative.

· Deduce variations.

### Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 25% increase

2. Calculate $\frac{1}{3}$ and $-\frac{1}{3}$. .....

3. Differentiate $x^2 - 5x + 1$. .....

4. Solve $(x-2)(x+1) = 0$. .....

5. Give the forbidden value of $\frac{x+1}{x}$. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate a polynomial and study a derivative.

### Investigation

For $f(x) = 2x^3 - 3x^2 - 12x + 5$, calculate $f'(x)$, factorise, study its sign and deduce the variations of $f$.

### Written summary

$(ax^3 + bx^2 + cx + d)' = 3ax^2 + 2bx + c$. The sign of the derivative gives the direction of variation of the function.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

### Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 5 + \frac{7}{x}$.

2. Calculate $f'(x)$ and study its sign.

3. Study on $]0, +\infty[$ the function $g(x) = x + \frac{25}{x}$.

**References in the textbook**

**Extensions:** Calao, pp. 94-96. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

**Calao textbook · p. 94, Exercise 13**

Let $f$ the function defined on $\mathbb{R}^*$ by: $f(x) = \frac{1}{x}$.

1. Calculate $f'(x)$.
2. Show that for every real $x$ non-zero: $f'(x) = \frac{-1}{x^2}$.
3. After studying the sign of $f'(x)$, draw the variation table of $f$.

**Independent work**

Redo practice exercise 3 with the following data: replace the initial value with 260 and the rate or common ratio with 6. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

## Session 29 — Complete study and graph

Reciprocal function

19/01/2027 • B : 2 h – C : 1 h

### Objectives of the session

· Carry out a complete study.

· Draw a coherent curve.

· Interpret an extremum.

### Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x + 4 - x + 2$. .....

2. Differentiate 3 $x^2 - 5x + 1$. .....

3. Solve $(x - 2)(x + 1) = 0$. .....

4. Give the forbidden value of $\dfrac{x+1}{x}$. .....

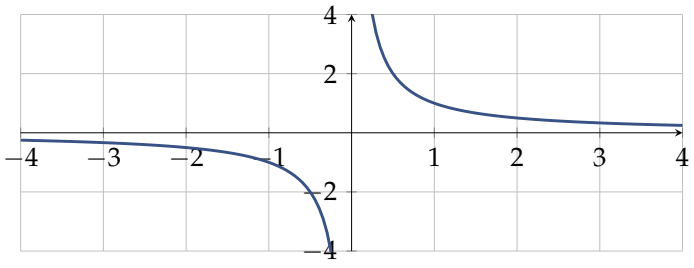
5. Calculate $\dfrac{1}{4}$ and $\cancel{-1}\ \dfrac{1}{4}$. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: carry out a complete study and draw a neat curve.

### Investigation

Study fully $g(x) = x + \dfrac{9}{x}$ on $]0, +\infty[$ : domain, derivative, sign, variations, minimum, behaviour at the bounds and sketch.




### Written summary

A full study includes: domain, derivative, sign, variations, limits or behaviour at the bounds, notable points and a consistent graph.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

**Practice · essentials**

1. Give the domain and the asymptotes of $f(x) = 6 + \frac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \frac{36}{x}$.

**References in the textbook**

**Extensions:** Calao, pp. 98-103. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

**Calao textbook · p. 98, Exercise 31**

A company specialising in industrial ovens can make between 20 and 80 of them depending on the years. Let $x$ the number of ovens manufactured annually.
The graph given in the textbook is that of a function $C$ defined on the interval $[20; 80]$. For all integer values of $x$, $C(x)$ is the average unit production cost (in €).

1. What is the average unit production cost when 25 ovens are produced, then when 70 ovens are produced?
2. Which production levels correspond to an average unit cost of 20,000 € ?
3. What is the minimum average unit production cost? What output level does it correspond to?

**Independent work**

Redo practice exercise 2 with the following data: replace the initial value with 265 and the rate or common ratio with 7. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

## Session 30 — Review and assessment

Reciprocal function

21/01/2027 • B : 1 h – C : 2 h

### Objectives of the session

· Use domain, derivative and sign.

· Write up a study.

· Correct methodically.

### Fluency · 6 to 8 minutes, without calculator

1. Calculate 5% of 100. .....
2. Solve $(x-2)(x+1)=0$. .....
3. Give the forbidden value of $\dfrac{x+1}{x}$. .....
4. Calculate $\dfrac{1}{5}$ and $-\dfrac{1}{5}$. .....
5. Differentiate 3 $x^2-5x+1$. .....

### Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: use domain, derivative and sign and write a study.

### Investigation

Correct a deliberately flawed study of $h(x)=x+\dfrac{4}{x}$ : stated domain $\mathbb{R}$, derivative $1+\dfrac{4}{x^2}$, table with no break at 0. Rewrite a complete solution.

### Written summary

Check the four items separately: domain, expression of the derivative, sign of the derivative, variations. An error at one step must not hide the following ones.

### Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

### Practice · essentials

1. Give the domain and the asymptotes of $f(x)=2+\dfrac{4}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0,+\infty[$ the function $g(x)=x+\dfrac{4}{x}$.

**References in the textbook**

**Extensions:** Calao, pp. 100-103. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

**Calao textbook · p. 98, Exercise 32**

Each week, a liquid detergent company estimates that the production cost (in euros) can be modelled by a function $C$ given by: $C(x) = x^2 + 60x + 121$, where $x$ is the volume of detergent (in m$^3$), $x \in [1;30]$.
Let $f$ the function representing the average cost of production per m$^3$ of liquid detergent.

1. Show that for every $x \in [1;30]$, $f(x) = x + 60 + \dfrac{121}{x}$.
2. Calculate $f'(x)$. Show that for every $x \in [1;30]$, $f'(x) = \dfrac{(x-11)(x+11)}{x^2}$.
3. Draw the variation table of $f$.
4. What is the minimum average production cost? For what quantity of detergent is it reached ?

**Independent work**

Redo practice exercise 3 with the following data: replace the initial value with 270 and the rate or common ratio with 2. Write four lines: data, formula, calculation, conclusion.

**Exit ticket · 3 minutes**

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.